EXPERIMENT 03

HEURISTIC EVALUATION

Website + Mobile App / Key task analysis

AIM

To evaluate a website and its mobile app for key user tasks against Nielsen's usability heuristics, rate the severity of each problem and recommend improvements.

SUBJECT

City Library Portal

Website (laptop) + Mobile app (Android)

PROCEDURE

01 PICK

Choose the website and app, then list 4 key user tasks.

02 WALK

Walk through every task alone and note each issue.

03 TAG

Map every issue to one of the 10 heuristics.

04 RATE

Score severity from 0 (none) to 4 (catastrophe).

05 MERGE

Combine group findings and write the fixes.

10 HEURISTICS // SEVERITY 0-4 // KEY TASKS // RECOMMENDATIONS

NIELSEN'S 10 HEURISTICS

H1

Visibility of system status

Keep users informed with timely feedback.

H2

Match with the real world

Speak the user's language, not jargon.

H3

User control and freedom

Offer clear undo and emergency exits.

H4

Consistency and standards

Same words and actions mean the same thing.

H5

Error prevention

Stop problems before they can happen.

H6

Recognition over recall

Make options visible, don't make users remember.

H7

Flexibility and efficiency

Give experts shortcuts and accelerators.

H8

Aesthetic and minimal design

Show only what is relevant right now.

H9

Help users recover from errors

Plain-language messages with a clear fix.

H10

Help and documentation

Searchable, short, task-focused help.

SEVERITY SCALE

0	Not a problem	Disagreement only
1	Cosmetic	Fix if time permits
2	Minor	Low-priority fix
3	Major	High-priority fix
4	Catastrophe	Fix before release

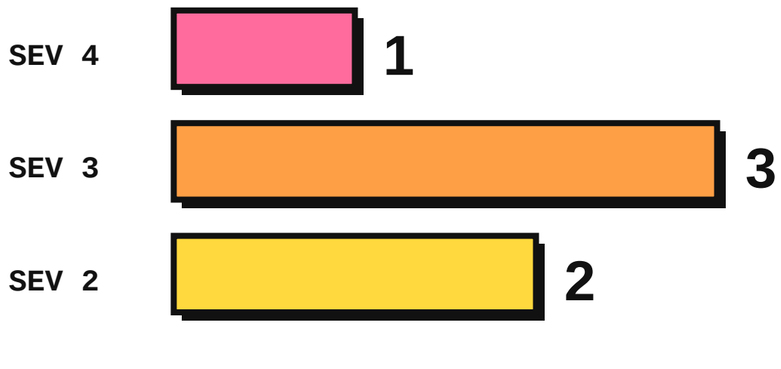
KEY TASKS + FINDINGS

ID	TASK	PLATFORM	FINDING	HEURISTIC	SEVERITY
T1	Search for a book	WEB	Results load with no spinner or progress cue; the page jumps when they appear.	H1 Status	2 / 4
T1	Search for a book	APP	Filter options hide behind an unlabeled icon, so users must remember it exists.	H6 Recognition	3 / 4
T2	Renew a loan	WEB	'Renew' sits beside 'Return' with identical styling and no confirmation step.	H5 Prevention	3 / 4
T3	Create an account	WEB	Error reads 'Invalid input (0x42)' and the faulty field is not highlighted.	H9 Recovery	4 / 4
T3	Create an account	APP	Form says 'Card No.' while the website calls the same thing 'Member ID'.	H4 Consistency	2 / 4
T4	Reserve a study room	APP	No back or cancel during booking; the only exit is closing the app.	H3 Control	3 / 4

Severity is judged on three factors: frequency of the problem, impact on the task, and persistence across uses.

SEVERITY + FIXES

PROBLEMS BY SEVERITY



2.8 / 4

average severity. 4 of 6 problems are rated 3 or above.

1 — Rewrite the sign-up errors

Highlight the faulty field and say what to fix in plain words, e.g. 'Card number must be 10 digits.'

H9 · SEV 4

2 — Separate Renew from Return

Give the two actions different colours and add a one-tap confirmation before renewing.

H5 · SEV 3

3 — Show filters and exits openly

Replace the icon with a labelled 'Filters' button and add Back / Cancel at every booking step.

H6 / H3 · SEV 3

4 — Add feedback, unify labels

Show a loading state on search and use one term ('Member ID') across web and app.

H1 / H4 · SEV 2

RESULT The heuristic evaluation of the library website and mobile app was completed. Six usability problems were found against six heuristics, four of them rated major or catastrophic. Prioritised fixes are documented for the redesign.